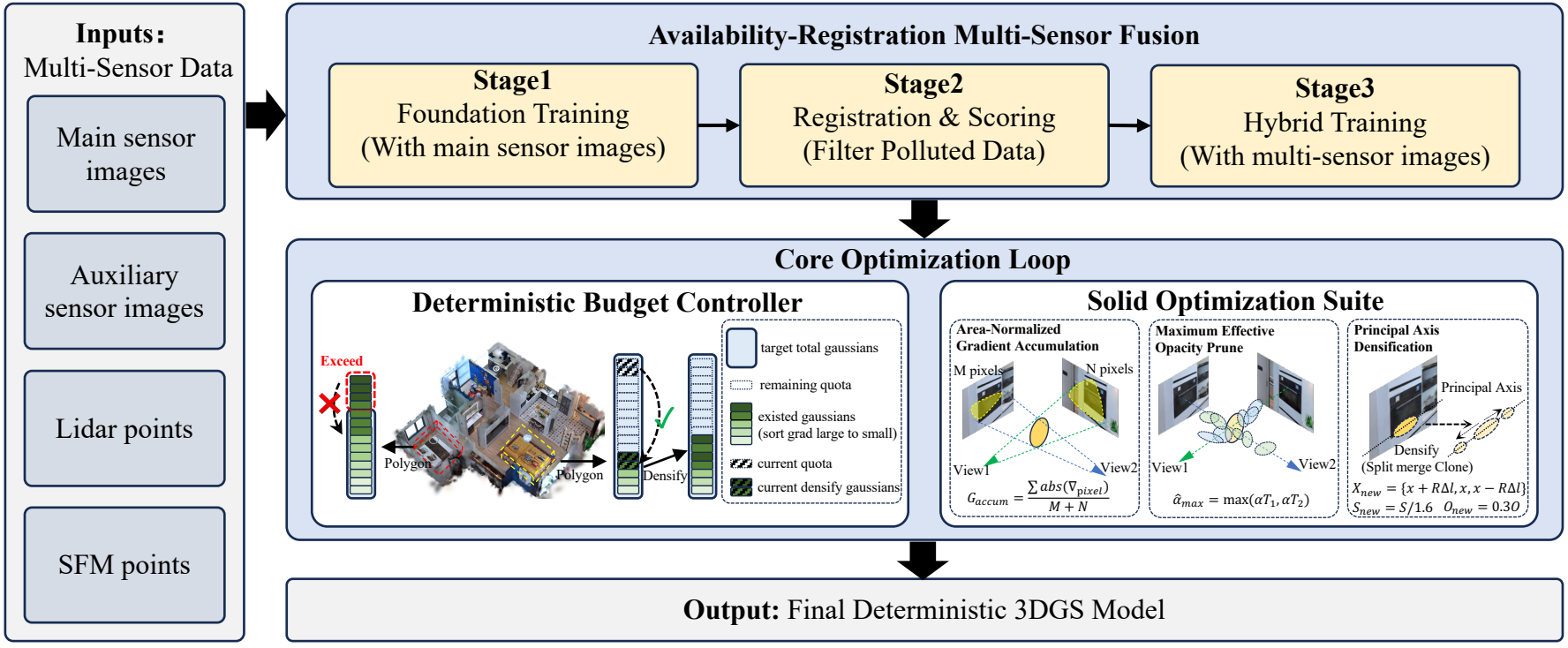
Inputs:
Multi-Sensor Data
Main sensor images
Auxiliary sensor images
Lidar points
SFM points
Availability-Registration Multi-Sensor Fusion
Stage1
Foundation Training
(With main sensor images)
Stage2
Registration & Scoring
(Filter Polluted Data)
Stage3
Hybrid Training
(With multi-sensor images)
Core Optimization Loop
Deterministic Budget Controller
Exceed
Polygon
Polygon
Densify
target total gaussians
remaining quota
existed gaussians
(sort grad large to small)
current quota
current densify gaussians
Solid Optimization Suite
Area-Normalized
Gradient Accumulation
M pixels
N pixels
View1
View2
$G_{accum} = \frac{\sum abs(\nabla_{pixel})}{M+N}$
Maximum Effective
Opacity Prune
View1
View2
$\hat{\alpha}_{max} = \max(\alpha T_1, \alpha T_2)$
Principal Axis
Densification
Principal Axis
Densify
(Split merge Clone)
$X_{new} = \{x + R\Delta l, x, x - R\Delta l\}$
$S_{new} = S/1.6 \quad O_{new} = 0.3O$
Output: Final Deterministic 3DGS Model